

ABG Interpretation in 6 Steps

pH, PaCO₂, HCO₃⁻, compensation, anion gap and oxygenation, with worked examples

NORMAL VALUES

Typical adult arterial values at sea level.

VALUE	NORMAL	NOTE
pH	7.35–7.45	Midpoint 7.40
PaCO ₂	35–45 mm Hg	Respiratory (acid)
HCO ₃ ⁻	22–26 mEq/L	Metabolic (base)
Base excess	–2 to +2 mEq/L	Metabolic only; negative = acidosis
PaO ₂	80–100 mm Hg	Room air; some labs 75–100
SaO ₂	95–100%	Room air
Anion gap	About 12	Lab ranges vary

SAFETY Sampling: Allen test before a radial stick; no air bubbles; analyze promptly; hold firm pressure, longer if anticoagulated; after O₂ or vent changes, wait per protocol.

THE 6 STEPS

1. pH: below 7.35 acidemia, above 7.45 alkalemia.
2. PaCO₂ and HCO₃⁻: mark each high, normal or low.
3. Primary disorder: the value that explains the pH (ROME).
4. Compensation: compare the other value with the expected range.
5. Metabolic acidosis? Anion gap, albumin correction, delta ratio.
6. Oxygenation: PaO₂, P/F ratio, A–a gradient.

EXAM TIP ROME: Respiratory Opposite, Metabolic Equal (pH versus PaCO₂ or HCO₃⁻). NCLEX stages: uncompensated (only the primary value off), partially compensated (both off, pH abnormal), fully compensated (pH normal; under 7.40 leans acid).

SAFETY Compensation seldom returns pH fully to normal. A normal pH with abnormal PaCO₂ and HCO₃⁻ is often two disorders at once.

EXPECTED COMPENSATION

Row 1 is Winter's formula. A value well outside the range suggests a second disorder.

PRIMARY	EXPECTED RESPONSE	COMPLETE IN
Metabolic acidosis	PaCO ₂ = 1.5 × HCO ₃ ⁻ + 8 (±2)	12–24 h
Metabolic alkalosis	PaCO ₂ up 0.6–0.7 per 1 HCO ₃ ⁻ rise; rarely over 55	24–36 h
Acute resp. acidosis	HCO ₃ ⁻ up 1 per 10 PaCO ₂ rise	Immediate
Chronic resp. acidosis	HCO ₃ ⁻ up about 4 per 10	3–5 days
Acute resp. alkalosis	HCO ₃ ⁻ down 2 per 10 PaCO ₂ fall	Immediate
Chronic resp. alkalosis	HCO ₃ ⁻ down 4–5 per 10	2–3 days

SAFETY DKA: a PaCO₂ above the Winter's range signals a tiring patient. Acute asthma: PaCO₂ runs low early; normal or above 45 signals impending respiratory failure. Escalate early.

ANION GAP AND DELTA RATIO

- Anion gap = Na⁺ – (Cl⁻ + HCO₃⁻)
- Corrected gap = gap + 2.5 × (4 – albumin in g/dL)
- Delta ratio = (corrected gap – 12) ÷ (24 – HCO₃⁻), a rough guide
- Ratio under 1: added normal-gap acidosis
- Ratio 1–2: pure high-gap acidosis
- Ratio over 2: added metabolic alkalosis

EXAM TIP Exams test which acidoses raise the gap: DKA, lactate and toxic alcohols do; diarrhea and large-volume saline do not. Kussmaul breathing is the classic compensation cue.

OXYGENATION

- P/F = PaO₂ ÷ FiO₂ (as a decimal)
- ARDS (intubated, PEEP 5 or more): mild over 200 to 300, moderate over 100 to 200, severe 100 or less
- SpO₂/FiO₂ 315 or less (if SpO₂ 97% or less) can stand in for P/F 300 (2024 definition)
- PAO₂ = FiO₂ × (760 – 47) – PaCO₂ ÷ 0.8 at sea level; room air about 150 – 1.25 × PaCO₂

- A–a gradient = PAO₂ – PaO₂; normal about age ÷ 4 + 4
- Low PaO₂ + normal A–a: hypoventilation or low inspired O₂ (altitude)
- Low PaO₂ + high A–a: lung problem (V/Q mismatch, shunt, diffusion)

COMMON CAUSES

DISORDER	COMMON CAUSES
Respiratory acidosis	Opioids, sedatives, COPD, asthma fatigue, neuromuscular weakness
Respiratory alkalosis	Pain, anxiety, hypoxemia, PE, early sepsis, pregnancy, salicylates
High-gap metabolic acidosis	Lactate, ketoacidosis, kidney failure, toxic alcohols, salicylates
Normal-gap metabolic acidosis	Diarrhea, large-volume saline, renal tubular acidosis
Metabolic alkalosis	Vomiting, NG suction, diuretics, hypokalemia

SAFETY Potassium: acidemia tends to shift K⁺ out of cells, alkalemia shifts it in. In DKA, serum K⁺ may be normal or high despite a total-body deficit, and it falls once insulin starts.

WORKED EXAMPLES

1. DKA, Kussmaul breathing: pH 7.22, PaCO₂ 25, HCO₃⁻ 10, Na⁺ 138, Cl⁻ 100, albumin 4.0
 - Metabolic acidosis; Winter's 23 (21–25), PaCO₂ 25 fits
 - Gap 138 – 110 = 28; delta ratio 16 ÷ 14 ≈ 1.1
 - Answer: high-gap metabolic acidosis, compensated as expected (NCLEX: partially compensated)
2. Opioid overdose, age 24, room air: pH 7.26, PaCO₂ 60, HCO₃⁻ 26, PaO₂ 68
 - Respiratory acidosis; acute rule predicts 26, matches
 - A–a: 150 – 75 – 68 = 7, normal for age
 - Answer: acute (uncompensated) respiratory acidosis; hypoxemia from hypoventilation
3. Septic shock, 5 L saline, ventilated, PEEP 10, FiO₂ 0.6: pH 7.29, PaCO₂ 30, HCO₃⁻ 14, Na⁺ 140, Cl⁻ 112, albumin 2.0, PaO₂ 90
 - Metabolic acidosis; Winter's 29 (27–31), PaCO₂ 30 fits
 - Corrected gap 14 + 5 = 19; delta ratio 7 ÷ 10 = 0.7
 - Answer: high-gap (likely lactate) plus normal-gap (saline) acidosis; P/F 150, moderate ARDS range